

Chapter 161 — Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy

EVOLUTION OF MINIMALLY INVASIVE LAPAROSCOPIC PROSTATECTOMY

- **Anatomic foundations (late 1970s–early 1980s):** cadaveric studies defined the **dorsal venous complex (DVC)**, the **neurovascular bundle (NVB; Walsh and Donker, 1982)** and the **striated urethral sphincter** → anatomic, nerve-sparing open radical prostatectomy with markedly reduced morbidity; it remained the standard for localised disease for over two decades.
- **First laparoscopic radical prostatectomy (LRP): Schuessler, 1997** → very long operating times and prolonged stay; cure appeared comparable to open surgery but **no advantage could be shown**, so LRP was not adopted.
- **Revival (2000): two French centres** published a stepwise, reproducible and teachable technique → acceptable operating times and positive margin rates; worldwide uptake followed, but **laparoscopic suturing of the vesicourethral anastomosis** remained the skill barrier.
- **Robotic assistance** was adopted largely to overcome that barrier. The **da Vinci Surgical System** became dominant in urology through:
 - **wristed (EndoWrist) instruments** reproducing the dexterity of the human wrist;
 - **10× magnified, three-dimensional stereo-endoscopic view**, versus the two-dimensional view of conventional laparoscopy.

DA VINCI PLATFORM GENERATIONS

Platform	Year	Distinguishing feature
First generation	2000 (United States launch)	Three arms: two instrument arms + one camera arm
da Vinci S	2006	High-definition image; fourth arm for grasping and retraction
da Vinci Si HD	2009	Dual surgeon consoles → two surgeons operate simultaneously; efficiency and training
da Vinci Xi HD	2014	Multiquadrant and simpler single-quadrant surgery; side docking ; no hip flexion needed to dock
da Vinci SP	2018	Single port : all instruments through one 2.5-cm cannula

- **Pattern across rows:** each generation added one capability — image quality, a fourth arm, a second console, docking flexibility, then a single incision.
- **Robotic-assisted laparoscopic radical prostatectomy (RALRP)** has grown rapidly since 2000 and is now the **dominant surgical approach in the United States** and in parts of Europe and Asia; **LRP remains common where robotic penetrance is low**.
- **Unresolved questions:** relative oncologic and functional merits versus open surgery, weighed against **equipment cost** and the **learning curve** for surgeon and team.

Concept — *The robot did not change the operation; it removed the laparoscopic suturing barrier, which is why RALRP spread where LRP had stalled.*

PATIENT SELECTION

Indications and Contraindications

- **Indications identical to open surgery:** clinically localised cancer — **cT1–T2**, or **cT3** still amenable to extirpation with a wide resection.
- **Staging imaging is risk-stratified** (2022 American Urological Association/American Society for Radiation Oncology [AUA/ASTRO] Clinically Localized Prostate Cancer guideline):
 - **High risk** → cross-sectional pelvic imaging (computed tomography [CT] or magnetic resonance imaging [MRI]) **and bone scan: obtain.**
 - **Unfavourable intermediate risk** → **consider** the same.
 - **Prostate-specific membrane antigen positron emission tomography (PSMA PET)-CT** → consider in high risk for metastasis with **negative conventional imaging.**

CONTRAINDICATIONS TO MINIMALLY INVASIVE PROSTATECTOMY

Absolute	Relative (technical challenge, not contraindication)
Uncorrectable bleeding diathesis	Neoadjuvant hormonal therapy (distorted planes)
Unfit for general anaesthesia (severe cardiopulmonary compromise)	Prior complex lower abdominal/pelvic surgery (partial colectomy, inguinal mesh herniorrhaphy)
	Morbid obesity; large prostate (>80 g); median lobe; prior transurethral resection of the prostate (TURP); salvage after radiotherapy or ablation

- **What the table tests:** only two things make the operation impossible — bleeding that cannot be corrected and inability to tolerate general anaesthesia. Everything else is **difficulty, to be avoided early in a surgeon's experience.**
- **Prior laparoscopic extraperitoneal mesh herniorrhaphy** → **prefer transperitoneal**; dense retropubic adhesions make extraperitoneal access to the space of Retzius difficult.

Morbid obesity.

- **Risks:** **compression nerve injury** (extra padding) and **respiratory compromise in steep Trendelenburg** (close communication with anaesthesia once pneumoperitoneum is established).
- **Technical constraints:** limited working space, trocar size and instrument length.
- **Modifications:** **extra-long (16 cm) trocars**; trocars **1–2 cm more cephalad** (better angle to the apex, less collision with the pubic symphysis); **divide the urachus more cephalad** (removes hanging fat); **lower pneumoperitoneum to 10 mm Hg** with deeper trocar insertion to reach the apex and anastomosis.
- **Bottom line:** experienced surgeons achieve adequate operative, functional and oncologic outcomes.

Large prostate (>80 g) and median lobe.

- **Challenges:** gland fills the pelvis → little working space and poor manoeuvrability; **median lobe** complicates posterior bladder neck division near the **ureteric orifices** and may need extensive bladder neck reconstruction; **wider DVC**, wider and more vascular **pedicles**; **NVB displaced posteriorly.**
- **Review preoperative imaging** (transrectal ultrasound, CT or MRI) to anticipate a large gland, median lobe or bladder stones.
- **Outcome:** longer operating time and more blood loss, but **long-term continence comparable** to smaller glands.

Prior TURP.

- **Identify the ureteric orifices before the posterior bladder neck dissection;** the prostate–bladder neck plane is often distorted.

- **Oncologic and functional outcomes may be worse** than in primary cases.

Salvage surgery (after radiation, brachytherapy, cryotherapy or high-intensity focused ultrasound [HIFU]).

- Feasible in selected patients, but **periprostatic fibrosis obliterates the prostate-rectum plane** → higher risk of **rectal injury**.
- Requires **full bowel preparation** and counselling on **rectal injury, possible colonic diversion, and higher rates of impotence and incontinence** than primary surgery.

Use of Preoperative Multiparametric Magnetic Resonance Imaging

- **Multiparametric MRI (mpMRI) for local staging: high specificity (0.88), low sensitivity (0.61).**
- **Reader dependent** — a dedicated reader improves prediction of adverse pathology.
- **Added to clinical risk tools → better prediction of extracapsular extension (ECE)** → sharper counselling on the balance of cancer control versus function.
- **Surgical planning:** guides the **extent of NVB preservation; capsular effacement** by a suspicious lesion → tailored wider resection or intraoperative frozen section → **fewer positive margins**, even in high risk.
- **Not proven necessary in every patient**, but review it when available.

Concept — *Selection turns on difficulty, not eligibility: indications are the same as open surgery, and nearly every adverse feature is a relative contraindication that experience overcomes.*

PREOPERATIVE PREPARATION

Patient Preparation

- **No routine formal bowel preparation;** magnesium citrate plus a Fleet enema is optional to empty the rectum for visibility and in case of rectal injury.
- **Prophylactic antibiotic before incision**, covering enteric gram-negative bacilli (e.g. **cefazolin**); **positive preoperative urine culture → targeted treatment.**
- **Low-dose aspirin may be continued;** other antithrombotics are withheld for the appropriate interval.

Informed Consent

- **Specific to minimally invasive surgery: conversion to open surgery; general anaesthesia is mandatory** (cannot be done under regional anaesthesia).
- **Shared with open surgery:** bleeding, transfusion, infection, impotence, incontinence, incisional hernia, lymphocele, injury to ureter, rectum, bladder or small bowel.
- The surgeon discloses **personal experience** and gives an **individualised forecast** of cancer control and function.
- **Baseline validated questionnaires: Sexual Health Inventory for Men (SHIM) and International Prostate Symptom Score (IPSS).**

Operating Room Personnel

- **Whole team trained** in the equipment, setup and steps: scrub technician, circulating nurse, assistant(s).
- **One skilled assistant usually suffices;** for RALRP the assistant must know basic laparoscopy **and robot setup and troubleshooting.**
- **Anaesthetist familiar with prolonged pneumoperitoneum** is vital.
- **Room layout (Fig. 161.1):** RALRP — separate **surgeon console**, one bedside assistant; LRP — surgeon at the table, first assistant, optional second assistant and optional **AESOP robotic camera holder.**

Patient Positioning

Sequence: general endotracheal anaesthesia → **arms tucked and padded** (egg-crate; protects **median and ulnar nerves**; Fig. 161.2) → **sequential compression devices** on → legs in **low lithotomy** or split-leg table (Xi side-docking: modified supine, "frog-leg") → **chest strapped** with tape and egg-crate against sliding → check chest compliance with anaesthesia → orogastric tube and urethral catheter.

- **Avoid fixed shoulder rests** → shoulder and **brachial plexus** compression in steep Trendelenburg.
- **Hip flexion:** older platforms needed slight flexion to dock; **Xi does not. Exaggerated flexion** → **femoral neuropraxia**.
- **Take a history of prior injuries** and pad hips, shoulders, knees and calves.

Anaesthesia Considerations

- Effects span musculoskeletal, intravascular, acid-base, ocular, respiratory and neurological systems → **work with a consistent anaesthesia team**, especially early.
- **Secure oximetry, blood pressure cuff and intravenous access before final positioning** — the arms are inaccessible afterwards.
- **Carbon dioxide (CO₂) pneumoperitoneum:** initial **vagal response, oliguria, hypercarbia** → uncorrected, **systemic acidosis**. Response: anaesthetist raises minute and tidal volumes; surgeon lowers insufflation pressure. Worse with the long operating times of early experience.
- **Fluid management:** urine output is unreliable (oliguria; bladder open) and excess urine obscures the field → typically **1.5–2 L crystalloid, mostly after the anastomosis**; further fluid matched to blood and insensible losses.
- **Steep Trendelenburg:** **facial, orbital, pulmonary and laryngeal oedema**; **raised intraocular pressure** (no apparent long-term consequence in healthy patients); risk of **corneal oedema and abrasion** → lubricate and protect eyes. **Marked facial or orbital oedema** → **suspect laryngeal oedema** → **consider delaying extubation**.

Concept — *Almost every preparation step answers one of two stresses — **prolonged CO₂ pneumoperitoneum** and **steep head-down tilt with inaccessible limbs** — so access, padding and communication must be settled before the patient is tipped.*

KEY POINTS: PATIENT SELECTION, INSTRUMENTATION AND PREOPERATIVE PREPARATION

- **Avoid complex anatomy early in the learning curve** (large gland, median lobe, morbid obesity, prior pelvic surgery, post-radiation, post-TURP) — none is an absolute contraindication.
- **mpMRI informs ECE risk** and so the balance of cancer control against NVB preservation.
- **A skilled bedside assistant** trained in the robot improves efficiency.
- Surgeon and anaesthetist anticipate **hypercarbia and acidosis, corneal oedema, raised intraocular pressure and neurapraxia**.

INSTRUMENTATION

- Depends on approach and, for RALRP, on a three- versus four-arm robot.
- **LRP camera control:** **AESOP 3000** robotic arm (hand, voice or foot control) or a human assistant.
- **RALRP (S, Si, Xi): four arms**, one being the stereo-endoscope. Start with a **0° lens**; **left arm** grasper (Maryland bipolar or plasma kinetic dissector), **right arm** curved monopolar scissors, **fourth arm ProGrasp** (large atraumatic retractor). The surgeon **toggles control between any two of the three working arms**.

BOX 161.1 SUGGESTED INSTRUMENTATION FOR LAPAROSCOPIC AND ROBOTIC-ASSISTED LAPAROSCOPIC RADICAL PROSTATECTOMY

Item	LRP	RALRP
Camera	10-mm 0° and 30° laparoscopes; AESOP 3000 (optional)	Robotic endoscope, 0° and 30°
Dissection / energy	Monopolar scissors; monopolar hook; bipolar forceps; ultrasonic shears ; Maryland dissector	EndoWrist Maryland bipolar or PK dissector; curved monopolar scissors
Retraction	Assistant	EndoWrist ProGrasp
Suturing	Laparoscopic needle drivers ×2	EndoWrist large needle drivers ×2
Access	Veress needle ; 5-mm trocars ×3; 12-mm trocars ×2	12-mm trocars ×2; 8-mm metal robotic trocars (×3 with fourth arm)
Other	Suction-irrigator; 20-Fr van Buren sound ; 18-Fr catheter	18-Fr catheter
Clips	Small and medium-large Hem-o-lok clips (both)	
DVC suture	0 polyglactin (GS21)	0 polydioxanone
Posterior reconstruction	2-0 polydioxanone (both)	
Anastomosis	3-0 poliglecaprone (Monocryl), double-armed (both)	

- **What the box shows:** the two kits differ in **camera control, retraction and access** (LRP relies on assistants and laparoscopic ports; RALRP on EndoWrist instruments and 8-mm robotic trocars). **Sutures and clips are almost identical** — the reconstruction is the same operation.
- **Trap:** the DVC suture differs — **polyglactin for LRP, polydioxanone for RALRP**.

SURGICAL TECHNIQUE**Robotic-Assisted Versus Pure Laparoscopic Approach**

- **Principles of dissection are the same** for LRP and RALRP; the description centres on RALRP using the multiport platform.
- **da Vinci = master-slave system** with three components: **patient-side cart, surgeon console, video cart**.
- **Bedside assistant and scrub technician:** dock and undock; the assistant gives suction-irrigation, retraction, passes sutures and changes instruments.
- **Surgeon console:** three-dimensional, 10×, high-definition view; full control of the camera and three instrument arms; finger-loop master controls replicate hand and wrist movements in real time.

da Vinci SP (single port).

- All instruments through **one 2.5-cm cannula**; can be seated in an access system (e.g. GelPOINT) so instruments "float" outside the skin and articulate almost at once → larger working range.
- **Learning curve even for experts: 7 degrees of freedom** achieved by a different mechanism from Xi.
- **Limited bedside assistance** (compact trocar) → solutions: **remotely operated suction-irrigation (ROSI)** controlled by surgeon foot pedal, or an extra assistant trocar ("**SP + 1**").
- Better dexterity and vision than laparoscopy with fewer incisions; **same dissection principles**.
- **Expert laparoscopists** may find the robot unnecessary; **for most surgeons** it eases the anastomosis and the **critical angles for NVB preservation**.

Transperitoneal Approach

- **Favoured by most surgeons:** more working space, familiar landmarks.

TRANSPERITONEAL ANTERIOR VERSUS TRANSPERITONEAL RETROVESICAL APPROACH

	Anterior (commonest)	Retrovesical (posterior)
First step	Enter the space of Retzius immediately	Dissect seminal vesicles and vasa behind the bladder , in the cul-de-sac
Seminal vesicles / vasa	Dissected after bladder neck division , from in front	Dissected first , lateral → medial toward the ejaculatory ducts
Best suited	Routine cases	Median lobe (lobe, urine and blood obscure vesicles from the front)

Abdominal Access, Insufflation, and Trocar Placement

- **Access:** Veress needle at the base of the umbilicus or **open Hasson** technique; consider the site and nature of prior surgery. **Left upper quadrant is often safe with midline scars.** The authors enter with a **visual obturator** after insufflation.
- **Adhesions:** an offset working laparoscope through one trocar clears space for further trocars; the robot then clears the rest.
- **Insufflation 12–15 mm Hg;** steep Trendelenburg; secondary trocars under vision.
- **RALRP configuration (Fig. 161.3A):** camera trocar (12 mm for S/Si; 8 mm for Xi) just below or above the umbilicus, **15–17 cm from the pubic symphysis;** **three 8-mm robotic trocars;** assistant **12-mm and 5-mm trocars on the patient's right.**
- **Camera:** surgeon moves it by foot pedal plus arm movements. **0° throughout** for most; some switch to **30° down** for the **bladder neck, NVBs and apex**, especially in obesity, large glands, prior TURP or a narrow pelvis. Xi flips 30° up/down from the console.
- **LRP configuration (Fig. 161.3B):** surgeon on the **patient's left** using the two **pararectus** trocars; one or two assistants use the lateral trocars; camera held by assistant or **AESOP.**

Extraperitoneal Approach

Sequence: **1.5-cm infraumbilical incision** → through anterior rectus sheath → finger dissection anterior to the posterior rectus sheath and peritoneum → **trocar-mounted balloon dilator** advanced to the pubis in the midline → about **500 mL of air** under 0° vision develops the space of Retzius (Fig. 161.4) → secondary trocars → proceed as for the anterior approach.

Pros and Cons of Extraperitoneal Versus Transperitoneal Approach

EXTRAPERITONEAL VERSUS TRANSPERITONEAL ACCESS

	Extraperitoneal	Transperitoneal
Working space	Smaller; collapses with vigorous suction (less so with high-flow insufflators such as AirSeal)	Larger; familiar landmarks
Bowel	Peritoneum is a barrier → less bowel injury, bowel does not fall into field; less ileus , earlier diet in some series	Bowel exposed
Trendelenburg	May avoid steep tilt → suits obesity or pulmonary compromise	Steep tilt needed
CO ₂ absorption	Higher → more hypercarbia and acidosis; higher minute volume	Lower
Prior extraperitoneal mesh hernia repair	Difficult (retropubic space obliterated)	Preferred

	Extraperitoneal	Transperitoneal
Preferred for	Extensive prior abdominal surgery; morbid obesity	Most surgeons, most cases

- **Evidence:** early retrospective series favoured extraperitoneal LRP for operating time (saved at access to Retzius), diet, stay and early continence; **most studies show little or no difference.**
- **Verdict: surgeon preference;** no consistent advantage; be proficient in both.
- **Trap:** the peritoneum-intact approach protects bowel but costs space and **absorbs more CO₂.**

Developing the Space of Retzius

Sequence: inspect pelvis (Fig. 161.5) → **sharply release sigmoid adhesions to the left pelvic side wall** → divide the **urachus high** and incise peritoneum **just lateral to the medial umbilical ligaments** (cauterise obliterated umbilical vessels) → **prevesical fatty alveolar tissue confirms the plane** → posterior-cephalad traction on the urachus; blunt and sharp dissection of the avascular retropubic space (Fig. 161.6) → lateral bladder mobilisation down to where the **medial umbilical ligament crosses the vas** (tension-free anastomosis later) → defat the anterior prostate; **bipolar coagulation of superficial DVC branches.**

- **Stay within the pelvic brim** → avoids **epigastric vessels (anterior)** and **iliac vessels (lateral).**
- **Landmarks now visible (Fig. 161.7):** anterior bladder and prostate, **puboprostatic ligaments, endopelvic fascia, pubis.**
- **Endopelvic fascia and puboprostatic ligaments divided sharply; levator fibres preserved** and swept bluntly off the prostate to expose the deep DVC and urethra at the apex. **Avoid electrocautery** (pelvic floor, external sphincter, NVBs).
- **Accessory pudendal arteries** run longitudinally on the anteromedial prostate; **preserve them** — in some men they are the **dominant arterial supply to the corpora cavernosa.**

Ligation of the Deep Dorsal Venous Complex

- **Pneumoperitoneum tamponades venous bleeding** → the profuse DVC haemorrhage of open surgery is less evident, even if the complex is entered.
- ProGrasp (or the LRP assistant) gives fixed cephalad retraction of prostate and bladder to expose DVC and apex.

METHODS OF DEEP DVC CONTROL

Method	Technique	Point
Suture ligation (standard)	0 polydioxanone or polyglactin, needle passed beneath the DVC and anterior to the urethra, as close to the pubis / as far from the apex as possible (Fig. 161.8); may need puboprostatic ligament division; optional second distal stitch to suspend DVC to the pubis	Distance from apex avoids iatrogenic apical entry
Linear stapler	Ligates and divides in one step	Speed
Delayed control	Leave DVC until apical dissection → divide sharply → oversew in vertical mattress ; briefly raise insufflation pressure	Preserves anterior sphincter support (puboprostatic ligaments)

Bladder Neck Identification and Transection

- **Challenge:** no tactile feedback for the prostate–bladder junction.

MANOEUVRES TO LOCATE THE BLADDER NECK

Cue	What it shows	Pitfall
1. Fat line	Prevesical fat ends where the fat-free anterior prostate begins	—
2. Catheter balloon traction	Repeated caudal traction marks the junction	Prior TURP: balloon falls into the defect → division too distal. Balloon off midline → suspect median lobe → divide more proximally
3. Bladder dome traction	Cephalad pull tents the bladder neck at its prostatic attachment	—
4. Instrument "pinch"	Bimanual "palpation" with two instrument tips confirms the margin	—

Transection sequence: anterior bladder neck divided horizontally in the midline until the catheter appears (**not too lateral** — bladder pedicle bleeding) → deflate balloon; catheter tip lifted anteriorly with **external countertraction at the meatus** to suspend the prostate → **inspect posterior neck for a median lobe and the ureteric orifices** → **posterior neck divided horizontally in the midline**, angled 45° downward (Fig. 161.9) to avoid both the prostate base and the posterior bladder wall.

- **Median lobe:** vertical mucosal drop-off = **no median lobe**; a mass effect = median lobe → deliver with ProGrasp; a **transverse or sagittal cystotomy** may be needed to see beneath a large lobe.
- **Prior TURP:** margin distorted; **ureteric orifices often very close to the posterior margin**.
- **No bladder neck sparing after TURP or with a median lobe;** a 30° down lens helps; **when in doubt, divide more proximally** → avoids a positive bladder neck margin.

Dissection of the Seminal Vesicles and Vasa Deferentia

- **Anterior approach:** vesicles and vasa identified in the gap between the divided bladder neck and prostate and freed from the NVBs; **hemoclips, not electrocautery** (Fig. 161.10).
- **Retrovesical approach:** first step, deep in the cul-de-sac; vasa dissected **lateral → medial** to the ejaculatory ducts; **vesicles lie lateral and slightly posterior to the distal vasa**, freed from the NVB with hemoclips, antegrade from tip to base, **without thermal energy** (Fig. 161.11).
- **Retrovesical is especially useful with a median lobe.**

Development of the Plane Between the Prostate and Rectum

- **Purpose:** avoid rectal injury; define the **medial border of the pedicles and NVBs**. Antegrade development is unfamiliar to open surgeons but adapts readily.
- **Technique (Fig. 161.12):** anterior traction on vasa and vesicles → **2–3-cm transverse incision in the posterior layer of Denonvilliers fascia about 0.5 cm below the vesicle bases** → blunt dissection toward the apex without electrocautery.
- **Denonvilliers fascia** = inferior extension of the peritoneal cul-de-sac between prostate and rectum.

POSTERIOR PLANE ACCORDING TO CANCER RISK

	Low risk of ECE, full nerve sparing	Higher risk of posterolateral ECE (examination or mpMRI)
Denonvilliers fascia	Preserved (left on rectum)	Excised with the specimen

	Low risk of ECE, full nerve sparing	Higher risk of posterolateral ECE (examination or mpMRI)
Plane	Between Denonvilliers fascia and prostatic fascia ; to the apex and laterally to the pedicles; 30° up lens helps apically	Between Denonvilliers fascia and anterior perirectal fascia (fascia propria)

- **Substantial bleeding → too close to the prostate**; if the plane is elusive, retry to one side. In the right plane blunt dissection runs smoothly. **Mobilise the rectum well laterally and distally** now to protect it later.

Prostatic Pedicle Control

METHODS OF PROSTATIC PEDICLE CONTROL

Method	Advantage	Drawback
Monopolar or bipolar electrocautery	Simple	Thermal spread damages the NVB → limit or avoid
Hem-o-lok clips + sharp athermal division (commonest)	No thermal energy	Needs a thinned, well-delineated pedicle to lock; clips near the anastomosis can migrate into the bladder
Temporary bulldog clamp, then suture after specimen removal	Athermal	Extra step

- **Clip migration → voiding symptoms, haematuria, recurrent urinary infection, bladder stones** (Fig. 161.13: eroded Hem-o-lok clip with stone causing gross haematuria) → **retrieve dropped clips; keep clips away from the anastomosis**.
- Correct pedicle division prevents both **positive margins** and **NVB damage**.

Preservation of the Neurovascular Bundles

- **Target: parasympathetic cavernous nerve fibres** in the NVB — the key and most demanding step in any approach.
- **Nerve distribution is diffuse and variable**: main confluence **posterolateral**; a posterior "hammock" of nerves; **anteromedial fibres** of debated significance → preserve as much periprostatic nerve tissue as possible.

PERIPROSTATIC FASCIAL DISSECTION PLANES (FIG. 161.14)

Plane	Lateral boundary	Posterior boundary	Use	Cost
Extrafascial	Outside levator fascia	Outside Denonvilliers fascia	Widest margin; most oncologically sound	High risk of complete erectile dysfunction if bilateral
Interfascial	Between prostatic and levator fascia	Between prostatic and Denonvilliers fascia	Preferred for presumed organ-confined disease	Balanced
Intrafascial	Between prostatic capsule and prostatic fascia	—	Full-thickness NVB preservation	More positive margins; needs selection and experience

- **Pattern**: moving inward trades margin for nerves. **Trap**: intrafascial is **not** the default nerve-sparing plane — **interfascial** is.
- **Landmark**: periprostatic vessels in the interfascial space are a visual surrogate for the nerves; a **dominant periprostatic artery** is identifiable in most cases.
- **Mechanisms of NVB injury**: incision, entrapment (suture or clip), thermal injury, traction.

Antegrade release sequence: incise **levator fascia sharply on the anteromedial midprostate** (Fig. 161.15) → enter the interfascial plane → blunt dissection along the **shiny white prostatic fascia**, sweeping the NVB posterolaterally → a visible **lateral NVB groove** forms (Fig. 161.16) → **hemoclips placed on the pedicle using the groove**, without entrapping nerves → divide pedicle.

- Releasing the NVB **before** mobilising the specimen helps avoid **traction injury**; wristed robotic instruments and motion scaling aid this. For LRP, **fine-tipped (0.8-mm) dissectors** have been described.
- **High anterior versus standard release:** uncertain whether high release preserves extra nerves or simply gives a handle to protect the **true cavernous nerves at 5 and 7 o'clock** without touching them.
- **Consensus: minimise, ideally avoid, thermal energy near the NVB** — cavernous fibres are highly heat sensitive (animal and human evidence).
- **Haemostasis:** NVB bleeding is usually minimal; small arteries or venous sinuses → **fine absorbable sutures with small, site-specific bites**.
- **Very large glands (>100 g):** exposure by assistant and fourth arm; consider a **30° down lens**.

Apical Dissection

- **The apex is the commonest site of positive margins**, and its dissection governs **continence and potency**.
- By now antegrade dissection has freed the base, lateral and posterior prostate; **DVC and urethra are divided last**.
- **Do not enter the anterior prostate** when dividing the DVC (iatrogenic positive margin). If the DVC stitch is too proximal, **remove it**, divide with a wider margin, then re-suture. Raise insufflation pressure transiently to limit bleeding.
- Then inspect the apex-urethra junction (Fig. 161.17): **preserve maximal urethral length** but recognise a **protruding anterior or posterior prostatic lip**.
- **Leaving a small rim of urethra on the apex** lowers apical positive margins **without harming continence**.
- **Sharp division, minimal electrocautery** → protects the external striated sphincter and NVBs.

Intraoperative Inspection of Prostate

- Before entrapment, **inspect the whole specimen surface**; excise extra tissue at any site of a close margin — **rarely needed with experience**.

Pelvic Lymphadenectomy

- Performed **after the prostate is freed and before entrapment** — bladder mobilisation already exposes the obturator fossa and iliac vessels (detail under Minimally Invasive Pelvic Lymph Node Dissection).

Entrapment of Specimens

- Prostate and nodes placed in a **10-mm entrapment sack** passed through the **12-mm assistant trocar**; left in the abdomen until the end.

Bladder Neck Reconstruction

- **Usually unnecessary:** the slightly larger bladder opening "parachutes" onto the urethra during the running anastomosis.
- **Needed with a large gland, median lobe or prior TURP** (bladder neck much larger than urethra) → **tennis-racquet closure**, posterior or anterior, with absorbable suture.
- **Ureteric orifices then lie at or near the posterior margin** → identify them first (intravenous **methylene blue or indigo carmine** if needed).

- **Protection options:** **imbricate the orifices** with interrupted 3-0 polyglactin or polydioxanone at **5 and 7 o'clock** (also narrows the neck), or **ureteric stents** removed at once or after healing.

Posterior Support of the Vesicourethral Anastomosis

- Prostatectomy divides **Denonvilliers fascia** and its confluence with the **posterior rhabdosphincter**.
- **Rocco stitch (posterior reconstruction):** reported **earlier return of continence**; other studies found **no benefit**. Proposed mechanisms: restored posterior support, better urethral coaptation, **less anastomotic tension**, longer functional sphincter.
- **Agreed minimum benefit:** shortens the bladder neck-urethra gap → **tension-free anastomosis**.
- **Technique (modified Rocco stitch, Fig. 161.18):** remnant **Denonvilliers fascia** and **posterior detrusor/bladder neck** → **posterior rhabdosphincter** beneath the urethra, **running 2-0 Monocryl**, before the anastomosis.
- **Anterior resuspension** of the anastomosis and distal bladder neck to the **arcus tendineus** restores anterior support and the vesicourethral angle.

Vesicourethral Anastomosis

- Hardest step of LRP; **the robot's wristed instruments greatly ease it**.
- **Van Velthoven running technique:** spreads tension across many points. Classically **two sutures tied end to end** (each 6–8 inches); now more often **double-armed barbed suture**.
- **No mucosal eversion sutures needed** (unlike historic open surgery) — the running closure gives **mucosa-to-mucosa apposition**.

Sequence (Fig. 161.19): start **posteriorly at 6 o'clock** → **preplace 2–3 throws each side** through urethra and bladder → **cinch by lifting each arm anteriorly** (assistant or ProGrasp holds tension on one arm) → run both arms anteriorly → **pass final catheter under vision** before closing → **irrigate to test for leak** → interrupted sutures if leaking.

- **Avoid incorporating the NVBs** in urethral bites.

Delivery of the Specimens and Exiting the Abdomen

Sequence: inspect for bleeding at low pressure (<10 mm Hg) → check nearby bowel for instrument injury → close **12-mm fascial sites** (e.g. Carter-Thomason device; **5- and 8-mm sites need no fascial closure**) → move sack string to the umbilical camera port → desufflate → **extract specimen intact via extended periumbilical incision** → close anterior rectus fascia → subcuticular absorbable skin closure.

Technical Modifications to Optimise Return of Urinary Control

- **Anterior compartment structures** — endopelvic fascia, **detrusor apron**, pubovesical and puboprostatic ligaments, DVC with overlying visceral endopelvic fascia — stabilise prostate, urethra and bladder and are thought to be part of the continence mechanism.
- Goal: **preserve rather than reconstruct** them → two techniques: **Retzius-sparing (RS)** and **hood**.

Retzius-Sparing Technique

- **Bocciardi approach, 2010:** **entirely posterior**, through the **rectovesical pouch (of Douglas)**; **no dissection in the anterior compartment** (Fig. 161.20).

Sequence: peritoneal incision in the pouch → isolate vasa and vesicles → separate Denonvilliers fascia antegrade along the posterolateral prostate to the apex (intra-, inter- or extrafascial as oncology dictates) → control pedicles and release NVBs antegrade, base to apex → continue anteriorly, **preserving the detrusor apron** → **bladder neck divided circumferentially, starting posteriorly** → dissect apically **beneath the**

detrusor apron and DVC (both preserved) → divide urethra → "upside-down" anastomosis: anterior first, posterior last.

- **Evidence (Cochrane systematic review of single-centre randomised trials):** earlier continence with RS at 1 week and 3 months; difference not clear by 6 months, equal by 12 months.
- **Erectile function at 12 months: no difference** (single-centre randomised trial).
- **Positive surgical margins: higher with RS** → use selectively; **locally advanced or anterior tumours are poor candidates** pending long-term oncologic data.

"Hood" Technique

- **Tewari group, 2021:** keeps a **familiar anterior approach** while preserving the anterior continence structures.
- **Four differences from the standard anterior approach:**
 - preservation of the **median umbilical ligaments**;
 - preservation of the **endopelvic fascia–puboprostatic ligament complex**;
 - **complete antegrade control of pedicles and release of NVBs**, posteriorly base to apex, then laterally;
 - **antero-apical dissection preserving the detrusor apron**, separated from the anterior fibromuscular stroma.
- Result: a **"hood"** of detrusor apron, arcus tendineus, puboprostatic ligaments and DVC over the **membranous urethra and external sphincter** (Fig. 161.21).
- **Early continence promising** in a small single-institution cohort; needs randomised trials.

STANDARD ANTERIOR VERSUS RETZIUS-SPARING VERSUS HOOD RALRP

	Standard anterior	Retzius-sparing	Hood
Route	Anterior, space of Retzius	Posterior, pouch of Douglas	Anterior
Anterior support (detrusor apron, puboprostatic ligaments, DVC)	Divided	Preserved (untouched)	Preserved (dissected off prostate)
Anastomosis	Posterior → anterior	Anterior → posterior ("upside-down")	Posterior → anterior
Continence	Reference	Faster to 3 months; equal at 12 months	Promising early data
Potency	Reference	No difference at 12 months	Not established
Margins	Reference	Higher positive margins	Not established
Evidence	—	Randomised trials; Cochrane review	Small cohort

Concept — Every step is a trade between **margin and function** made at a named boundary — bladder neck, Denonvilliers fascia, fascial plane, apex — and the robotic technique protects function by being **antegrade, athermal and plane-precise** rather than by removing less cancer.

KEY POINTS: TECHNIQUE OF LRP AND RALRP

- **Transperitoneal versus extraperitoneal: surgeon preference**; no significant advantage.
- The robot **eases suturing and dissection** for surgeons without advanced laparoscopic skills.
- **Bladder neck margin:** found by physical manoeuvres and visual cues.
- **Define median lobe presence after anterior, before posterior, bladder neck division.**

- **Full posterior prostate-rectum mobilisation** prevents rectal injury and defines the medial pedicles and NVBs.
- **Minimise thermal energy** at pedicles and NVBs.
- **Interfascial dissection** for organ-confined cancer when potency is wanted.
- **Anticipate a protruding apical lip** to avoid an apical positive margin.
- **Bladder neck reconstruction** for size mismatch — prior TURP, median lobe, large gland.
- **Running continuous suture** is the most efficient anastomosis.
- **RS and hood techniques** show promise for earlier continence.

POSTOPERATIVE MANAGEMENT

Enhanced Recovery After Robotic Prostatectomy

- **Enhanced Recovery After Surgery (ERAS)**: multimodal, patient-centred — **education, nutrition, analgesia, early mobilisation**; increasingly standard after RALRP and a driver of **same-day discharge**.

Pain Management

Sequence: preoperative education on expected pain → **infiltrate all incisions with long-acting local anaesthetic** (bupivacaine) → **ketorolac 15–30 mg intravenously near the end** (unless non-steroidal anti-inflammatory drugs [NSAIDs] contraindicated) → **regular paracetamol (acetaminophen) + NSAID for 24–48 h**, then as needed → **opioids only for inadequate control**, short-acting and brief.

- Non-pharmacological: early ambulation, relaxation techniques.

Diet

- **Mild ileus** occasionally, resolving in **24–48 h**; **opioids worsen it**.
- **Oral fluids and diet within 4–8 h**; counsel on constipation — minimise opioids, use stool softeners.

Deep Venous Thrombosis Prophylaxis

- **Pneumatic compression** from before incision, continued postoperatively; **subcutaneous heparin 1–2 h before surgery** may be given until ambulant. **Thromboembolism is very uncommon**.

Same Day Discharge

- **Same-day discharge (SDD)** is feasible and safe in **selected patients, experienced surgeons, high-volume centres**; enabled by ERAS.
- **Success improved by less peritoneal irritation**: **ultra-low pneumoperitoneum (as low as 6 mm Hg)** or the **extraperitoneal approach**.
- Needs counselling and a structured follow-up plan; benefits — quicker return to activity, lower cost.

Pelvic Drain

- **Not routinely indicated** after a watertight running anastomosis. If used: through an **8-mm robotic port site** (no extra stab); drains unexpected urine or lymph; usually **removed day 1–2** if the patient is still in hospital.

Urethral Catheter

CATHETER DURATION BY SCENARIO

Scenario	Duration	Cystogram
Routine, watertight anastomosis	About 1 week (the 2 weeks of open surgery unnecessary); not earlier than 5 days	Not required
Extensive bladder neck reconstruction (very large gland, large median lobe, post-TURP) or doubtful anastomosis (post-radiation)	At least 10 days	Yes, before removal
Salvage prostatectomy	2–3 weeks	Strongly consider
Extravasation	Longer, until spontaneous healing	Repeat to confirm

- **Trap — early removal (day 3–4): no more radiographic leaks, but significantly more acute retention** needing recatheterisation → avoid removal before day 5.
- Extravasation delays continence but **does not affect long-term continence**.
- **Suprapubic catheter with an anastomotic splint** avoids a urethral catheter, with good early results.

Return to Normal Activity

- Catheter out at 1 week → light restriction for a further week → **all normal activity by 4 weeks** → **strenuous exercise at 6 weeks**. A clear advantage over open surgery.

Role of Continence Physiotherapy

- **Preoperative pelvic floor therapy helps:** exercises (especially **external sphincter contraction**) are learnt more easily before surgery than after, when inhibition, sensory change, leakage and pain interfere.
- **Postoperative physiotherapist-guided pelvic floor muscle training shortens time to continence** → access to a specialist physiotherapist is essential.
- **Start pelvic floor exercises no sooner than 3 days after catheter removal** — earlier exercise may cause perineal pain.

Role of Penile Rehabilitation

- **Post-prostatectomy erectile dysfunction (ED) has three causes:** neural injury, vascular injury, corporal smooth muscle damage.
- **Recovery depends on:** nerve-sparing technique and extent, age, baseline erectile function, surgeon experience.
- **Phosphodiesterase type 5 inhibitors (PDE5i)** are the core: protective to endothelium and smooth muscle.

MEMORIAL SLOAN-KETTERING CANCER CENTER (MSKCC) PENILE REHABILITATION PROTOCOL

Phase	Regimen
2 weeks before → 2 weeks after surgery	Low-dose PDE5i nightly
Weeks 3–6	Low-dose PDE5i 6 nights/week + maximum-dose PDE5i once weekly → classify responder vs failure
Responders	Continue PDE5i
Failures, from week 6	Intracavernosal injection (ICI) twice weekly + low-dose PDE5i on other nights

Phase	Regimen
From 12 months	Monthly maximum-dose PDE5i challenge; stop ICI once natural or PDE5i response returns

- **Trap: three randomised placebo-controlled trials** — PDE5i helps **on-demand** erections, but **early (rehabilitative) PDE5i does not improve later drug response**.
- **ICI** aids recovery and is used for PDE5i non-responders.
- **Vacuum erection device (VED)**: main rehabilitation benefit is **preserving penile length**.

Concept — *Recovery after RALRP is paced by the anastomosis, not the wound: catheter, drain, physiotherapy and discharge timings all follow from how secure the vesicourethral join is, while erectile recovery follows the nerves and is only supported, not created, by rehabilitation.*

PERIOPERATIVE OUTCOMES

- **Early single-institution reports** suggested superior potency and continence with RALRP — not borne out by controlled data.
- **Landmark trial** — Yaxley et al., 2016 (randomised, one open surgeon vs one robotic surgeon): equivalent continence, potency and early oncologic outcomes at 3 months; **RALRP favoured for blood loss** (mean about 1400 mL open vs 300 mL robotic) and **length of stay**.
- **Scandinavian surgeon-level comparisons**: outcomes depend chiefly on the **individual surgeon's skill and experience**; better with **high-volume surgeons at high-volume centres**.
- **Barrier to uptake: cost** — capital outlay and limited-use instruments.
- **Real advantages of RALRP**: minimally invasive recovery, **faster return to activity**, at least equivalent oncologic and functional results, and a **wider pool of surgical candidates** (prior mesh repair, older comorbid men, prior abdominal or pelvic surgery) who might otherwise be steered to external-beam radiotherapy.

Concept — *The surgeon, not the platform, is the dominant variable: randomised data show parity on cancer and function, so the robot's case rests on recovery and reach.*

ONCOLOGIC OUTCOMES

- **Measures**: positive surgical margin (PSM) rate and biochemical recurrence (BCR) — prostate-specific antigen (PSA) rising above 0.2 ng/mL after becoming undetectable.
- **Meta-analysis**: no difference in PSM between radical retropubic prostatectomy (RRP) and RALRP; mean RALRP PSM about 15% overall, about 9% in pathologically localised disease; 7-year BCR-free survival about 80%.
- **Cancer-specific survival needs 15–20 years** of follow-up → not yet defined for RALRP.
- **PSM as surrogate of surgical quality**: lower in high-volume centres; **varies between surgeons even in centres of excellence**.
- **Tumour factors drive margins, independent of technique**: Gleason ≥ 7 , perineural invasion, higher stage.
- **Trap** — **do not raise PSM**: obesity/high body mass index, large prostate, prior abdominal surgery.
- **Training**: early BCR higher for **open surgeons adopting RALRP** than for **fellowship-trained robotic surgeons**; BCR risk stays **stable once the learning curve is passed**.

Concept — *Margins measure the tumour first and the surgeon second; the approach itself contributes nothing once volume and training are accounted for.*

FUNCTIONAL OUTCOMES

Urinary Continence

PASSIVE VERSUS ACTIVE CONTINENCE

Passive	Active
Smooth muscle of bladder neck, prostatic urethra and urethra (Fig. 161.22)	External striated sphincter: horseshoe-shaped skeletal muscle at the apex (Fig. 161.23)
Suspensory system: endopelvic fascia, detrusor apron, pubovesical and puboprostatic ligaments, DVC with visceral endopelvic fascia	Fast-twitch fibres; contraction angulates the urethra
—	Pudendal nerve via Alcock's canal; branches swing from beneath the levator ani into the sphincter at the apex (Fig. 161.24) → injured by excessive dissection, heat or suturing there

- **Reporting problem:** continence definitions vary widely ("**no pad**" vs "**safety pad**") → incontinence rates vary widely.
- **High-volume experienced surgeons:** continence at 1 year is **excellent (no-pad rates in the mid-90s %)** and **comparable to RRP**.
- **Persistent mild incontinence** → suburethral sling or adjustable continence therapy (ProACT).
- **Persistent severe incontinence (<2% of men)** → artificial urinary sphincter.

Sexual Function

- **Before Walsh and Donker (1982)**, impotence after radical prostatectomy was the norm.
- **NVB composition:** sympathetic (hypogastric nerve, T12–L1) joins parasympathetic (S2–S4) just below the prostate base; a **ganglion posterolateral to the capsule below the base** gives the cavernous nerves, usually in their own fascial sheath. Nerves to prostate, vesicles, rectum and levator ani run in adjacent compartments (Fig. 161.25).
- **Course:** posterolateral to the prostate, anterolateral to the rectum, outside the capsule; **3–5 mm from the capsule** → **most men with localised cancer can be offered nerve sparing**.
- **At the apex** the cavernous nerves swing medially and anteriorly to about **3 and 9 o'clock at the mid-urethra** → **the apex is where erectile nerves are most often damaged** (Fig. 161.26).
- **Apical scarring** from extensive or repeated (especially transperineal) biopsies raises risk to the nerves.
- **Palpable extracapsular disease** → wider dissection, sacrificing one or both NVBs.
- **Nerve-sparing decision** now combines **Gleason grade, clinical stage, number and percentage of positive cores** with **mpMRI** (effacement or breach of a Prostate Imaging–Reporting and Data System [PI-RADS] lesion at the capsule).
- **Reporting problems:** many men **≥60 have pre-existing ED**; **younger men (40s–early 50s) recover better**; many series ignore PDE5i or ICI use; **recovery continues up to 5 years**, so a **12-month end point may be too early**.
- **Surgeon-dependent:** nerve preservation varies between surgeons.
- **Meta-analysis:** ED after nerve-sparing RALRP varies widely at 12 and 24 months, reflecting non-standard definitions, technique and rehabilitation.

- **High-volume experts:** potency (erections sufficient for intercourse in **>50% of attempts**) in **up to 90%** of men without prior ED at 1 year.

Concept — *Continence and potency both fail at the apex: the pudendal branches to the sphincter and the cavernous nerves converge there, which is why apical technique governs function.*

KEY POINTS: POSTOPERATIVE MANAGEMENT AND PERIOPERATIVE OUTCOMES

- **Most discharged within 24 h; normal activity in 3–4 weeks.**
- **Oncologic equivalence** robotic vs open; PSM lower with high-volume surgeons in high-volume centres.
- **Continence at 1 year excellent** in experienced hands; **potency varies** with age, baseline function, definitions, technique and rehabilitation, but is better with experienced surgeons.

COMPLICATIONS

Complications Related to Patient Positioning

- **History of musculoskeletal problems** → patient-specific padding.
- **Protect cervical, axillary, brachial and radial nerves;** pad arms and hands; care of the **brachial plexus** if arms extended.
- **No shoulder rests;** Yellowfins stirrups protect the legs.
- **Trendelenburg tilt 22–25°;** prolonged tilt → **ocular oedema**, occasionally **cerebral irritation**.
- **Excessive hip flexion** → **femoral nerve injury;** not needed with Xi.

Bleeding and Transfusion

- **Transfusion 1–2%;** **blood loss dramatically lower than open** (pneumoperitoneum compresses veins; vision identifies bleeders). In the Yaxley trial, **1400 mL open vs 300 mL robotic**.

Visceral Injury

- **Rectal, sigmoid and small bowel injuries:** rare but serious → intraoperative general surgical consultation.
- **Recognised rectal injury, minimal/no spillage** → **primary repair without colostomy:** multilayer closure + omental or peritoneal flap interposed between rectum and anastomosis.
- **Unrecognised rectal injury** → **rectovesical fistula** → may need **colostomy and later fistula repair**.
- **Small bowel injury:** often **outside the camera's view** (instrument exchanges by the assistant) or during adhesiolysis; unrecognised → **delayed acute abdomen and peritonitis** → **immediate repair and drainage**.

Ureteral Injury

THREE SITES OF URETERIC INJURY DURING RALRP

Step	Mechanism	Prevention / management
Vas dissection (posterior approach)	Ureter lies just posterior to the vas	Stay on the vasal adventitia
Trigone / posterior bladder neck	Prior TURP or large median lobe → orifices at the posterior margin → caught in the anastomosis	Imbricate orifices, or robotically placed double-pigtail stent (retrograde over a wire via assistant port; confirm on plain abdominal radiograph [KUB]; remove cystoscopically at about 1 month)
Proximal PLND	Ureter crosses the iliac vessels	Minimise thermal energy at the proximal packet

Thromboembolic Complications

- **AUA Best Practice Statement:** routine intermittent pneumatic compression for laparoscopic and robotic urological surgery; **pharmacological prophylaxis only with multiple risk factors** (obesity, advanced age, malignancy, immobility, prior deep vein thrombosis [DVT]).
- **Incidence as low as 0.5%** — faster mobilisation, and **Trendelenburg reduces lower-limb venous stasis** compared with open surgery.
- **Two-way link with pelvic collections:** DVT → anticoagulate and image the pelvis (CT or ultrasound) for a lymphocele, haematoma or urinoma compressing the external iliac vein; significant lymphocele or haematoma → low threshold for duplex ultrasound for occult ipsilateral DVT.

Anastomotic Complications

Urine Leak

- Can cause **significant ileus**. No drain → interventional radiology percutaneous drain; prolonged catheterisation; cystogram to confirm resolution.
- In skilled hands **rare, small, self-limiting, no long-term effect on continence**; major disruption may need **reoperation** and worsens continence.

Bladder Neck Stricture

- **Open RRP: up to 16%; RALRP: <2%** — attributed to precise **mucosa-to-mucosa apposition**.
- **Management ladder:** dilation or transurethral incision + catheter → rarely, for recalcitrant strictures, posterior urethroplasty with anastomotic revision.

Equipment Malfunction

- **Critical robotic failure about 0.4% (<0.5%)** → cancellation before starting, or **conversion to laparoscopic or open**; commonest failing parts: **robotic arms and optical system**.

COMPLICATION BENCHMARKS AND MANAGEMENT

Complication	Rate	Key management
Transfusion	1–2%	—
Thromboembolism	As low as 0.5%	Mechanical prophylaxis routine; drugs if multiple risk factors
Bladder neck stricture	<2% (open up to 16%)	Dilation / incision → urethroplasty
Robotic failure	<0.5%	Convert or cancel
Rectal injury (recognised)	Rare; higher in salvage	Primary multilayer repair + flap
Rectal injury (unrecognised)	Rare	Fistula → colostomy + repair
Urine leak	Uncommon	Drain, longer catheter, cystogram

Concept — *The robotic complication profile is one of fewer but less visible injuries: bleeding and strictures fall with magnified vision, while the dangerous events are those outside the camera's field — bowel injury, positioning neuropathy — that must be anticipated rather than seen.*

KEY POINTS: COMPLICATIONS OF RALRP

- **Positioning:** pad every compression point; respect pre-existing conditions.
- **Transfusion <2%** — a clear advantage over open surgery.
- **Rectal injury rare, usually repairable when recognised; commoner in salvage.** Small bowel injury is usually **off-screen** (instruments, trocars, adhesiolysis); consult general surgery for any bowel injury.
- **Urine leak:** longer catheter, usually resolves; **cystogram** to confirm.

- **Bladder neck stricture <2%.**
- **Critical machine failure <0.5%.**

SALVAGE ROBOTIC-ASSISTED RADICAL PROSTATECTOMY

- **Indication:** local failure after **external-beam radiotherapy (EBRT)**, brachytherapy, HIFU or cryotherapy.
- Historically avoided (open) because of **incontinence, ED, bladder neck stenosis and rectal injury** — rectal injury **above 15%** at some centres.
- **Robotic vision allows precise dissection through fibrosis**; salvage RALRP is **feasible and safe** in small series but with **worse function** than primary RALRP.
- **Work-up:** biopsy-proven local recurrence + imaging, **preferably PSMA PET**, to exclude metastases.
- **High-volume centres, experienced robotic surgeons.**

Salvage Prostatectomy Technique

- **After EBRT:** pelvic side wall and Retzius harder to develop; **desmoplastic reaction around the iliac vessels**. Planes usually **preserved after brachytherapy, cryotherapy or HIFU**.
- **Anterior:** incise endopelvic fascia to outline the prostate; **initial DVC suture often unnecessary** (apex hard to define; little back-bleeding after radiation).
- **Bladder neck:** find by exposing the lateral prostate and the **"pinch" manoeuvre**; neck may be **pale and thickened**.
- **Complete seminal vesicle excision** — invasion is commoner in recurrent disease.
- **Posterior plane is the danger zone:** rectal injury occurs posterior to Denonvilliers fascia → **full mechanical bowel preparation**; patient antegrade dissection of the perirectal fat plane, sharp and blunt. **Brachytherapy seeds** often met here (or anywhere).

Rectal check and repair sequence: inspect rectum after removal → **air bubble test** for a suspected occult injury → if no gross spillage and tissue adequate: **secure two-layer closure + omentum or regional flap interposed** → **any doubt: intraoperative colorectal consultation, consider diverting colostomy**.

- **Anastomosis:** same running technique, but **higher leak and later stricture risk** (poor vascularity) → **catheter 2–3 weeks; cystogram strongly advised** before removal.
- **Continence:** only **33–80% regain reasonable continence**.
- **Potency:** nerve sparing rarely feasible → **usually impotent**; many already have ED from prior treatment → counsel realistically.

PRIMARY VERSUS SALVAGE RALRP

	Primary	Salvage
Bowel preparation	Not routine	Full mechanical
DVC	Suture ligated	Initial ligation often unnecessary
Seminal vesicles	Dissected	Completely excised
Rectal injury risk	Rare	Higher (>15% historically, open)
Catheter	About 1 week	2–3 weeks + cystogram
Continence	Excellent at 1 year	33–80%

	Primary	Salvage
Potency	Nerve sparing usual	Usually impotent

Concept — *Salvage surgery is primary surgery through obliterated planes: every modification — bowel preparation, rectal testing, longer catheterisation — follows from radiation-induced fibrosis and poor vascularity.*

MINIMALLY INVASIVE PELVIC LYMPH NODE DISSECTION

Indications

- **Debated:** template extent, value of extended dissection, and any **therapeutic** benefit.

PELVIC LYMPH NODE DISSECTION (PLND) TEMPLATES

Template	Nodal groups	Share of prostatic drainage captured
Limited	Obturator only	—
Standard (sPLND)	Obturator + external iliac	~38%
Extended (ePLND)	Obturator + external iliac + internal iliac	~63%
+ Common iliac	ePLND + common iliac	~75%
Super-extended	ePLND + common iliac + presacral	—

- **What the table tests:** each step outward adds a nodal station; the standard template **misses most drainage** → **when PLND is done, use the extended template** (higher node yield and detection of metastases).
- **Therapeutic value:** conflicting effect on biochemical and clinical recurrence; **no improvement in cancer-specific or overall survival; more operative time, blood loss, stay and complications as extent increases.** Still justified as the **best staging tool**.
- **Low-risk disease:** nodal metastases in <5% → PLND may be omitted.

GUIDELINE INDICATIONS FOR STAGING PLND

	AUA (2018)	European Association of Urology (EAU, 2021)	AUA/ASTRO (2022)
Trigger	Unfavourable intermediate or high risk	Nomogram risk of nodal metastasis >5%	Risk-stratified, using preoperative nomograms; balance benefit against complications
Template	Extended when performed		

- **Caveat:** Briganti and MSKCC nomograms predate mpMRI and PSMA PET and assume **systematic (not targeted) biopsy** → who benefits from staging PLND is uncertain.
- **Benefits of finding node-positive disease:** refined risk stratification, prognostic counselling, selective secondary therapy.

Surgical Technique

- Laparoscopic or robotic; **rarely a stand-alone staging operation** now — usually with prostatectomy. Stand-alone: same trocars but **one assistant port**.

Sequence: vertical peritoneal incision just lateral to the medial umbilical ligament, from the iliac bifurcation to the pubis (Fig. 161.27; beware the ureter proximally; simplified if the bladder is already mobilised) → **separate the packet from the external iliac vein** → medial retraction; blunt dissection of the avascular plane to the side wall, point cautery only (Fig. 161.28) → extend **proximally to the bifurcation, distally to the pubis** → **identify and protect the obturator nerve and vessels** → clip and divide distally → retract cranially, free

from obturator nerve and vessels → **clip proximally, avoiding the obturator nerve and the accessory obturator vein** (Fig. 161.29) → extend to template → extract as one packet via 12-mm trocar or in the sack.

EXTENDED PLND BOUNDARIES

Boundary	Landmark
Proximal	2 cm above the common iliac bifurcation , where the ureter crosses
Distal	Node of Cloquet
Lateral	Genitofemoral nerve
Medial	Bladder wall
Content	All obturator fossa tissue above and below the obturator nerve ; tissue medial and lateral to the internal iliac artery (presacral in super-extended)

- **No valid evidence that open PLND beats laparoscopic**; an extended template can be cleared completely either way.
- **Node count is a poor quality surrogate** — **pathological processing** matters as much as surgical extent.
- **Clip visible lymphatic channels** → fewer lymphoceles.

Complications

- **Overall complication rate after PLND is appreciable (about one in seven).**
- **Vascular injury** (skeletonised iliac vessels): small venotomy or arteriotomy → **fine polypropylene suture**; major → **rapid open conversion**. Transfusion-requiring injury **well under 1%**.
- **Obturator nerve** (entrapment, cautery, transection) → **medial thigh numbness and weak ipsilateral adduction**. Clip on nerve → **remove it**; transection → **primary repair with fine polypropylene** for partial recovery.
- **Ureter**: uncommon; at risk proximally where it crosses the common iliac vessels.

Lymphocele — the commonest complication.

- **Radiological lymphocele in up to 60%**; clinically significant **2–5%**.
- **Trap: transperitoneal access is not protective** — fluid can still loculate.
- **Mechanism**: the bladder adheres to and **walls off the lymphadenectomy fossa** → **peritoneal interposition flap** keeps a window open to the peritoneum → significant reduction in clinically significant lymphocele (externally validated).
- **Asymptomatic** → **no treatment**.
- **Presentation**: classically **ipsilateral DVT** (external iliac vein compression), but **more often new or worse irritative voiding symptoms and pelvic pressure** (bladder compression); may become **infected**.
- **Management ladder**: percutaneous drain → recurrence: **sclerosant** or **laparoscopic fenestration and marsupialisation**.

Concept — *PLND is justified by staging, not cure: once done, it should be extended, and its morbidity is dominated by lymph that cannot drain — hence clips, flaps and windows.*

KEY POINTS: MINIMALLY INVASIVE PLND AND COMPLICATIONS

- **Who benefits from staging PLND remains debated.**
- **If performed, use the extended template.**
- **Clip lymphatics** to reduce lymphocele.
- **Transperitoneal approach does not prevent lymphocele.**

- **Peritoneal interposition flap** may reduce clinically significant lymphocele.
- **Symptomatic lymphocele:** bladder compression (irritative symptoms, pelvic pressure) or iliac vein compression (ipsilateral DVT).

FUTURE OF ROBOTIC PROSTATECTOMY

- **Expiry of key first-generation da Vinci patents (2019)** → new console platforms: **Medtronic Hugo**, **CMR Surgical Versius**, **Johnson & Johnson Ottava** and others.
- **Differentiators: modularity, flexible vs rigid instruments, multi- vs single-port, haptic feedback.**
Competition may **lower cost and widen global access.**
- **Artificial intelligence (AI):** pattern recognition in large datasets, real-time decision support, **performance assessment and outcome prediction**, individualised training feedback, eventually **semi-autonomous or autonomous functions.**
- **AI hurdles: volume of high-quality surgical data; data sharing, privacy and ethics;** safe response to **unpredictable intraoperative events.**
- **Training:** build new platforms into curricula; **graduated basic and procedure-specific simulation** (console, troubleshooting, foot-pedal coordination) **before live surgery**, modelled on **aviation; virtual and augmented reality** promising but resource-intensive and hard to validate.

SUMMARY

- **RALRP is the operation of choice** for localised and, increasingly, **locally advanced** disease; open prostatectomy is now less common in the United States. **LRP persists where robots are cost-prohibitive.**
- **23-hour and same-day discharge** → faster recovery than open surgery; robotic surgery has **widened surgical candidacy** and shown promise in the **salvage** setting.
- **Cost remains the main constraint;** new vendors may lower it. The future is tied to **AI** (with data, privacy and ethical challenges) and to **simulation and virtual-reality training.**

Concept — *The chapter's argument is **equivalence plus access**: RALRP matches open surgery on cancer and function, wins on recovery, and its future depends on cost and training rather than on further proof of efficacy.*